Class Test 4 – Green's Theorem

Multivariable Calculus

Time: 20 minutes **Max Marks:** 10

Maintain academic integrity. Cheating will not be tolerated.

1. Use Green's Theorem to evaluate

$$\oint_C (x^2 y\,dx + xy^2\,dy)$$

where C is the positively oriented boundary of the rectangle with vertices

$$(0,0),\ (2,0),\ (2,1),\ (0,1).$$

(4 marks)

2. Verify Green's Theorem for the vector field

$$\mathbf{F}(x,y) = (-y)\mathbf{i} + x\mathbf{j}$$

over the region R bounded by the circle

$$x^2 + y^2 = 4$$

oriented counterclockwise. (6 marks)

Green's Theorem: $\oint_C \mathbf{F}\cdot d\mathbf{r} = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA.$